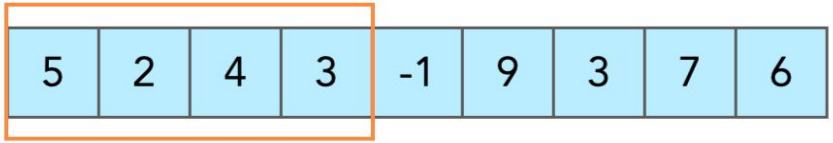
Sliding Window



Sliding window —> —>

Lecture Flow

- Pre-requisites
- Introduction and Basic Concepts
- One Dimensional problems
- Variants
- Complexity Analysis
- Applications of sliding window
- Two Dimensional problems
- Pitfalls
- Practice Questions

Pre-requisites

1. Arrays
2. Dictionary/Maps
3. Sets/Hash Sets
4. Two Pointers

Motivation Problem

Consider the problem of finding the maximum sum of 3 consecutive elements in an array.

How can we solve this problem ?

Can we just calculate the sum for each sub array size 3 and select the maximum sum?

Time complexity

- We are calculating each subarray sum on every iteration
- We have n iteration and k element
- So the Time complexity is : **O(n*k)**

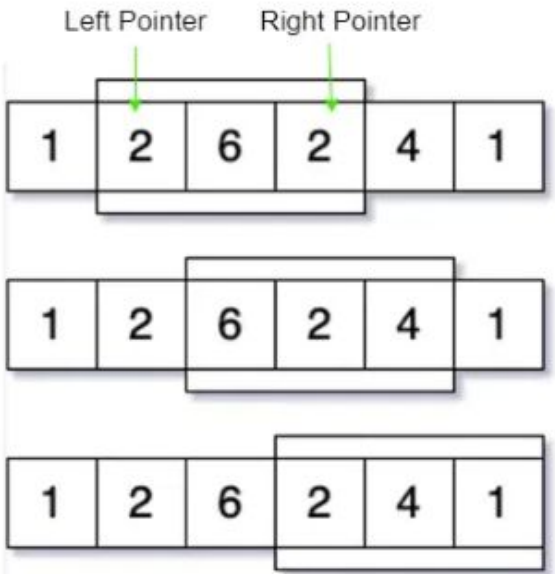
Can we do better ?

Introduction

- **Sliding Window** is a method used to efficiently solve problems that involve defining a window in the input data (arrays or strings)

- Then moving that window across the data to perform some operation within the window

- It Unlocks solutions for finding max/min values, counting elements, and pattern detection



Basic Concepts

The sliding window technique consists of two key components: **window size** and **window movement**.

- **window size** - the number of elements in the array or string that the window encompasses.

- **window movement** - the number of elements by which the window moves after each step.

One Dimensional problems

One-dimensional problems can be solved using the sliding window technique by creating a window of fixed size that moves over the array, one element at a time.

One Dimensional problems Cont.

Consider the problem of finding the maximum sum of k consecutive elements in an array.

How can we use sliding window to solve this problem ?

One Dimensional Sliding Window Simulation

2	1	4	3	0

K = 2

One Dimensional Sliding Window Simulation

Step: 1

2	1	4	3	0

maximum = 2+1 = 3

K = 2

One Dimensional Sliding Window Simulation

Step: 2

2	1	4	3	0

maximum = 1+4 = 5

K = 2

One Dimensional Sliding Window Simulation

Step: 3

2	1	4	3	0

maximum = 4+3 = 7

K = 2

One Dimensional Sliding Window Simulation

Step: 4

2	1	4	3	0

maximum = 3+0 = 3

Maximum sum of two items from this array is 7 and the pair is {4,3}.

K = 2

Variants

There are several types of sliding window:

1. Fixed Window Length **k**
2. Dynamic Sliding Window

Variant I

Fixed Size Sliding Window: Optimization

In this technique, a window of fixed size is moved over the input data, and some operation is performed on the data within the window.

For k = 3:

[a, b, c, d, e, f, g, h

[a, b, c]

[b, c, d]

[c, d, e]

[d, e, f]

[e, f, g]

[f, g, h]

Fixed Size Sliding Window: Optimization

Example: Given an array of integers, find the maximum sum of any contiguous subarray of size 3.

Optimization

- What is the difference between those two subarrays ?
- Can we construct the second one from the first one?
- If so how can we do that ?

[3,1,4,2,3,0]

[3,1,4,2,3,0]

Fixed Size Sliding Window: Optimization

```
max_sum = 0
```

```
[3,1,4,2,3,0]
```

Fixed Size Sliding Window: Optimization

max_sum = 8

[3,1,4,2,3,0]

Fixed Size Sliding Window: Optimization

max_sum = 8

[3,1,4,2,3,0]

A2SV

Fixed Size Sliding Window: Optimization

max_sum = 9

[3,1,4,2,3,0]

A2SV

Fixed Size Sliding Window: Optimization

max_sum = 9

[3,1,4,2,3,0]

A2SV

Practice Question

Problem Link

Implementation

```python
def findMaxAverage(nums, k):
    window_sum = 0
    max_sum = float('-inf')

    # Step 1: Compute the sum of the first k elements (initial window)
    for i in range(k):
        window_sum += nums[i]

    max_sum = window_sum
    left = 0
    # Step 2: Slide the window across the array
    for right in range(k, len(nums)):
        window_sum += nums[right]  # Add the new element entering the window
        window_sum -= nums[left]   # Remove the old element leaving the window

        max_sum = max(max_sum, window_sum)
        left += 1

    return max_sum / k
```

Time and space complexity?

Fixed-size sliding window

- Time complexity: **O(n)**
- Space complexity: **O(1)**

Variant II

Dynamic Sliding Window

In this technique, the size of the window is **not** fixed, and it can be increased or decreased based on certain conditions.

For example, given a string, find the longest substring with unique characters.

Dynamic Sliding Window

```
s = "abcbad"
```

```
longest = 0
```

Dynamic Sliding Window

s = "abcbad"

longest = 1

Dynamic Sliding Window

s = "abcbad"

longest = 2

Dynamic Sliding Window

s = "abcbad"

longest = 3

Dynamic Sliding Window

s = "abcbad"

```
longest = 3  // duplicate found, shrink window
```

Dynamic Sliding Window

s = "abcbad"

```
longest = 3  // duplicate found, shrink window
```

Dynamic Sliding Window

s = "abcbad"

longest = 3

A2SV

Dynamic Sliding Window

s = "abcbad"

longest = 3

Dynamic Sliding Window

s = "abcbad"

longest = 4

Longest substring without repeating character

Problem Link

Implementation

```python
class Solution:
    def lengthOfLongestSubstring(self, s: str) -> int:
        left, right = 0, 0
        adict = defaultdict(int)
        max_len = 0
        for right in range(len(s)):
            adict[s[right]] += 1
            # print(adict)
            while left < len(s) and len(adict) != (right-left+1):
                adict[s[left]] -= 1
                if adict[s[left]] == 0:
                    del adict[s[left]]

                left += 1

            max_len = max(max_len, right-left+1)


        return max_len
```

Time and space complexity?

Dynamic sliding window

- Time complexity: **O(n)**
- Space complexity: **O(1)**

Dynamic sliding window

Let's look at another question on dynamic sliding window.

[Problem Link]

General Template I

```
def dynamicwindowatmost(self, nums: str) -> int:
    n=len(nums)
    ans=0
    Window = # initialize
    left=0
    for right in range(n):
        Window = # update the window with right
        while window is invalid:
            Window = # remove the left part
            left += 1  # update left pointer

        ans= # update the answer

    return ans
```

General Template II

```
def dynamicwindowatleast(self, nums: str) -> int:
    n=len(nums)
    ans=0
    window= # initialize
    left=0
    for right in range(n):
        Window = # update the window with right
        while window is valid:
            ans= # update the answer
            Window = # remove the left part
            left += 1  # update left pointer


    return ans
```

Applications of sliding window

Sliding Window Problems

Sliding Window for Optimization

- Used when optimizing some metric (min/max value, longest/shortest condition, etc.).
- Example: Longest substring with at most k distinct characters.

Sliding Window for Counting

- Used when the problem involves counting occurrences of elements within a window.
- Example: Counting anagrams of a string in another string.

Sliding Window: Counting

In this technique, a sliding window is used to count the number of occurrences of a particular item in the input data.

For example, given two strings s and p, find the number of anagrams of p in s.

Sliding Window: Counting

```
s = "cbadabac"
p = "abc"
```

Sliding Window: Counting

anagrams = 0

s = "cbadabac"

target = { a:1, b:1, c:1 } `// represent p as a counter`

Sliding Window: Counting

anagrams = 1

s = "cbadabac"

target = { a:1, b:1, c:1 }

window = { a:1, b:1, c:1 } // represent each window as a counter and compare to target

Sliding Window: Counting

anagrams = 1

s = "cbadabac"

target = { a:1, b:1, c:1 }

window = { a:1, b:1, d:1 }

Sliding Window: Counting

anagrams = 1

s = "cbadabac"

target = { a:1, b:1, c:1 }

window = { a:2, d:1 }

Sliding Window: Counting

anagrams = 1

s = "cbadabac"

target = { a:1, b:1, c:1 }

window = { a:1, b:1, d:1 }

Sliding Window: Counting

anagrams = 1

s = "cbadabac"

target = { a:1, b:1, c:1 }

window = { a:2, b:1 }

Sliding Window: Counting

anagrams = 2

s = "cbadabac"

target = { a:1, b:1, c:1 }

window = { a:1, b:1, c:1 }

Practice Question

Problem Link

Implementation

```python
from collections import Counter
def findAnagrams(s, p):
        result = []
        target_counts = Counter(p)
        window_counts = Counter(s[:len(p) - 1])

        for i in range(len(p) - 1, len(s)):
            curr_char = s[i]
            window_counts[curr_char] += 1
            if window_counts == target_counts:
                result.append(i - len(p) + 1)

            window_counts[s[i - len(p) + 1]] -= 1
            if window_counts[s[i - len(p) + 1]] == 0:
                del window_counts[s[i - len(p) + 1]]

        return result
```

Time and space complexity?

Fixed-size sliding window

- Time complexity: **O(n)**
- Space complexity: **O(1)**

Two Dimensional problems

Two-dimensional problems can be solved using the sliding window technique by moving a window over the matrix , one row or column at a time.

Applications of 2D Sliding Window:

1. Image Processing
2. Pattern Matching
3. Matrix-based DP Problems - Problems like "Maximum Sum Submatrix of Size k × k".

Two Dimensional problems Cont.

Consider the problem of finding the maximum sum of a submatrix of size k x k in a matrix.

How can we solve this problem using sliding window ?

Two Dimensional problems Cont.

To solve this problem using the sliding window technique,

- We would create a window of size k x k and
- Move it over the matrix, summing up the elements in each window and
- Keeping track of the maximum sum we have seen so far.

Two Dimensional Sliding Window Simulation

5	3	1
2	4	2
0	3	1

Two Dimensional Sliding Window Simulation

Step: 1

5	3	1
2	4	2
0	3	1

maximum = 5+3+2+4 = 14

Two Dimensional Sliding Window Simulation

Step: 2

5	3	1
2	4	2
0	3	1

maximum = 3+1+4+2 = 10

Two Dimensional Sliding Window Simulation

Step: 3

5	3	1
2	4	2
0	3	1

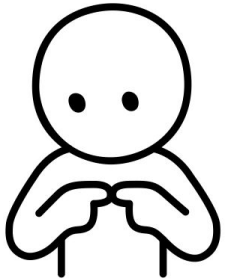
maximum = 2+4+0+3 = 9

Two Dimensional Sliding Window Simulation

Step: 4

5	3	1
2	4	2
0	3	1

maximum = 4+2+3+1 = 10



Common Pitfalls

Pitfalls

There are several common pitfalls when using sliding windows:

1. Off-by-one error
2. Not considering all possible cases
3. Not handling edge cases
4. Inefficient window updates
5. Not understanding the problem requirements

Complexity Analysis

- Fixed-size sliding window
  - Time complexity: **O(n)**
  - Space complexity: **O(1)**
- Dynamic sliding window
  - Time complexity: **O(n)**
  - Space complexity: **O(1)**
- Counting sliding window
  - Time complexity: **O(n)**
  - Space complexity: **O(1)**

Applications of sliding window technique

- Maximum Sum Subarray of Size K
- Distinct Elements in Every Window of Size K
- Maximum Consecutive Ones (with At Most K Flips)
- Anagram Search in a String
- Smallest Subarray with Sum ≥ K
- Longest Subarray with Sum ≤ K

Checkpoint

Practice Questions

- Minimum Size Subarray Sum
- Longest Repeating Character Replacement
- Permutation in String
- Smallest Range Covering Elements from K Lists
- Shortest Subarray with Sum at Least K
- Fruit Into Baskets
- Longest Turbulent Subarray
- Get Equal Substrings Within Budget
- Minimum Window Substring
- Codeforce's practice problems

A2SV

Quote Of The Day

"Your big opportunity may be right where you are now. Open your eyes, and you'll find it in your windows."

- Napoleon Hill